

Linear Algebra

In-Depth Study Notes

Chapter 2, Part 2 — Rank, Existence & Uniqueness, Invertible Matrices, Determinants

How This Continues Part 1

This picks up exactly where *LinearAlgebra_Ch2_Part1_Notes.pdf* left off. Part 1 covered §2.1–2.4 (what a linear system is, Gaussian elimination, row reduced echelon form, elementary matrices). This file, **Part 2**, finishes the chapter: §2.5–2.8 – rank, the existence/uniqueness theorem, invertible matrices, and determinants (including Cramer’s rule). Practice problems are embedded directly after each section this time (orange boxes); full worked solutions to all of them are in the companion *LinearAlgebra_Ch2_Part2_Solutions.pdf*. Colour code:

- **Blue** = original *Definition*, restated precisely.
- **Purple** = original *Theorem / Result*.
- **Green** = “In Plain English.”
- **Grey** = a fully worked example.
- **Orange** = a practice problem for you to try.
- **Coloured bold text** = a jargon word, explained on first use.

Chapter 3 (Vector Spaces) picks up after this, whenever you’re ready for it.

Section Map

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1 §2.5 Rank of a Matrix

You have already been computing rank informally every time you row-reduced a matrix in Part 1 – this section just makes it official.

Definitions 2.5.1 & 2.5.2 – Consistency and Row Rank

A linear system is **consistent** if it has at least one solution, and **inconsistent** if it has none. The **row rank** of a matrix A is the number of nonzero rows in its row reduced form.

Definition 2.5.6 – Rank

The **rank** of a matrix A , written $\text{rank}(A)$, is its row rank: the number of nonzero rows once A is row-reduced.

In Plain English

Rank measures “how much real information” a matrix actually contains, after you’ve thrown away rows that are just combinations of other rows. Row-reduce, count the surviving nonzero rows – that count never changes no matter which sequence of row operations you use, which is exactly why it’s safe to call it a single number attached to A .

Worked Example – computing rank

Let $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{pmatrix}$. Eliminate using row 1 ($R_{21}(-2)$, $R_{31}(-1)$):

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_{32}(-1)} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Two nonzero rows survive, so $\text{rank}(A) = 2$ (out of a possible maximum of 3) – row 3 of the original matrix carried no information beyond what rows 1 and 2 already gave.

Theorem 2.5.7

If $\text{rank}(A) = r$, there exist invertible matrices P, Q (each a product of elementary matrices) such that

$$PAQ = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}.$$

In Plain English

This says every matrix, no matter how messy, can be simplified down to “an identity block plus zeros” just by using row operations (P) and column operations (Q) together. The size of that identity block is exactly the rank – this is arguably the cleanest possible definition of what rank “physically is.”

Problems 2.5.1 – 2.5.5

2.5.1 Find $\text{rank} \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{pmatrix}$.

2.5.2 Find $\text{rank} \begin{pmatrix} 1 & 1 & 2 \\ 2 & 2 & 4 \\ 1 & 0 & 1 \end{pmatrix}$.

2.5.3 Give an example of a 3×3 matrix with rank exactly 1, and one with rank exactly 0.

2.5.4 Can a 2×5 matrix have rank 3? Explain why or why not, in terms of the maximum possible number of nonzero rows after row reduction.

2.5.5 (*Challenge*) If $\text{rank}(A) = r$ for an $m \times n$ matrix A , what is the largest possible value of r in terms of m and n ? Justify briefly.

2 §2.6 Existence and Uniqueness of Solutions

This is the payoff of everything in §2.3–2.5: a complete, no-exceptions answer to “how many solutions does $Ax = b$ have?”

Theorem 2.6.1 – Existence and Non-existence

Let A be $m \times n$, and consider $Ax = b$. Let $r = \text{rank}(A)$ and $r_a = \text{rank}([A \ b])$. Exactly one of the following holds:

- (1) If $r_a = r < n$: **infinitely many solutions** (there are $n - r$ free variables).
- (2) If $r_a = r = n$: a **unique solution**.
- (3) If $r < r_a$: **no solution**.

In Plain English

Notice $r_a \geq r$ is always true (adding a column to A can only keep the rank the same or increase it by exactly 1) – so there are only ever these three cases, never a fourth. This single theorem is the rigorous version of the “three outcomes” picture from §2.1: rank tells you exactly which of the three you’re in, with zero geometric guessing required.

Worked Example – revisiting Part 1’s three systems

Recall the three systems solved by hand in §2.3. Computing ranks confirms Theorem 2.6.1 predicts the outcome every time, *before* doing any back-substitution:

- **Unique-solution system** ($x+y+z=6$, $2x+y-z=1$, $x-y+2z=5$): $\text{rank}(A) = 3 = \text{rank}([A \ b]) = n \Rightarrow$ unique solution. ✓
- **Infinite-solutions system** ($x+y+z=4$, $2x+2y+2z=8$, $x-y+z=2$): $\text{rank}(A) = 2 = \text{rank}([A \ b]) < 3 = n \Rightarrow$ infinitely many solutions. ✓
- **No-solution system** (same as above but with $2x+2y+2z=9$): $\text{rank}(A) = 2 < 3 = \text{rank}([A \ b]) \Rightarrow$ no solution. ✓

The rank of A alone doesn’t change between the last two systems – it’s purely the extra column b pushing $\text{rank}([A \ b])$ up to 3 that flips the answer from “infinite” to “none.”

Corollary 2.6.3

The homogeneous system $Ax = 0$ (where A is $m \times n$) has a **non-trivial** solution if and only if $\text{rank}(A) < n$.

In Plain English

For a homogeneous system, $b = 0$ so $[A \ 0]$ always has the same rank as A (appending a zero column never raises rank) – case (3) of Theorem 2.6.1 can never happen. So you're always in case (1) or (2): either $\text{rank}(A) < n$ (free variables exist, giving nontrivial solutions alongside the trivial one), or $\text{rank}(A) = n$ (only the trivial solution $x = 0$).

Proposition 2.6.4

The following two statements can never both be true for the same matrix A : (1) $Ax = b$ has a unique solution for *every* b ; (2) $Ax = 0$ has a non-trivial solution.

Problems 2.6.1 – 2.6.5

- 2.6.1** For $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ and $b = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$, compute $\text{rank}(A)$ and $\text{rank}([A \ b])$, then state how many solutions $Ax = b$ has.
- 2.6.2** Repeat Problem 1 with $b = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$ instead.
- 2.6.3** For a 4×4 system with $\text{rank}(A) = 4$, how many solutions does $Ax = b$ have, for *any* choice of b ? Explain using Theorem 2.6.1.
- 2.6.4** Does $x - y + 2z = 0$, $2x - 2y + 4z = 0$ have a non-trivial solution? Use Corollary 2.6.3 (compute the rank of the coefficient matrix first).
- 2.6.5** (*Challenge*) Explain, using Proposition 2.6.4, why a *non-square* matrix A (more unknowns than equations) can never have “ $Ax = b$ uniquely solvable for every b .”

3 §2.7 Invertible Matrices**Definition 2.7.1 – Inverse**

Let A be a square matrix of order n . B is a **left inverse** of A if $BA = I_n$; C is a **right inverse** if $AC = I_n$. A is **invertible** if there is a matrix B with $AB = BA = I_n$.

Lemma 2.7.2 – The Inverse Is Unique

If $AB = I_n$ and $CA = I_n$, then $B = C$.

In Plain English – a one-line proof worth remembering

$C = CI_n = C(AB) = (CA)B = I_n B = B$. That's the entire proof – just insert AB (which equals I_n) in one spot and CA (which also equals I_n) in another, and associativity does the rest. Because left and right inverses always agree when both exist, we're justified in talking about *the* inverse, denoted A^{-1} , rather than “an” inverse.

Theorem 2.7.4 – Basic Properties

$(A^{-1})^{-1} = A$; $(AB)^{-1} = B^{-1}A^{-1}$ (note the order flips!); and $(A^t)^{-1} = (A^{-1})^t$.

In Plain English

The flipped order in $(AB)^{-1} = B^{-1}A^{-1}$ is the same “socks and shoes” rule from everyday life: to undo “put on socks, then shoes,” you must take off shoes first, then socks – undoing a sequence of operations reverses their order.

Definition 2.7.6 & Theorem 2.7.7 – Full Rank and Invertibility

A square matrix A of order n has **full rank** if $\text{rank}(A) = n$. For a square A of order n , the following are **all equivalent**:

- (1) A is invertible.
- (2) A has full rank.
- (3) A is row-equivalent to I_n .
- (4) A is a product of elementary matrices.

Theorem 2.7.8 & Theorem 2.7.10 – More Equivalences

If $AB = I_n$ for *some* B (just a right inverse, nothing assumed about the left side), then A is automatically invertible – and the same is true starting from a left inverse alone. Also, for square A of order n , these are equivalent to the four above:

- (5) $Ax = 0$ has *only* the trivial solution $x = 0$.
- (6) $Ax = b$ has a (unique) solution for *every* $b \in \mathbb{R}^n$.

In Plain English – the single most useful theorem in the chapter

This is a genuine “good characterisation”: six seemingly different questions (Is A invertible? Does it have full rank? Can I row-reduce it to I ? Is it built from elementary matrices? Does $Ax = 0$ only have the obvious solution? Can I always solve $Ax = b$?) all have the *same* yes/no answer for a given square matrix. In practice this means: whichever of these six is easiest to check for your specific matrix, check that one – the other five come along for free. It also finally explains why elementary matrices from §2.4 mattered so much: they are, by definition, invertible (each has an obvious inverse – undo the row operation), and this theorem says *every* invertible matrix is secretly built from them.

Gauss–Jordan Method for Computing A^{-1}

Row-reduce the block matrix $[A \mid I_n]$ using Gauss–Jordan elimination. If the left block becomes I_n , the right block is A^{-1} . If the left block ever develops an all-zero row before reaching I_n , A is not invertible.

Worked Example – computing an inverse

Let $A = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$.

$$\begin{aligned}
 & \left[\begin{array}{ccc|ccc} 2 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_1(1/2)} \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \\
 & \xrightarrow{R_{21}(-1)} \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & -\frac{1}{2} & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_{32}(-1)} \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & -\frac{1}{2} & 1 & 0 \\ 0 & 0 & \frac{3}{2} & \frac{1}{2} & -1 & 1 \end{array} \right] \\
 & \xrightarrow{R_3(2/3)} \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \end{array} \right] \xrightarrow{\begin{matrix} R_{13}(-1/2) \\ R_{23}(1/2) \end{matrix}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & -\frac{1}{3} \\ 0 & 1 & 0 & -\frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 1 & \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \end{array} \right]
 \end{aligned}$$

So $A^{-1} = \begin{pmatrix} 1/3 & 1/3 & -1/3 \\ -1/3 & 2/3 & 1/3 \\ 1/3 & -2/3 & 2/3 \end{pmatrix}$. (We'll double-check this against the determinant/adjoint formula in §2.8, and reuse this exact A there.)

Problems 2.7.1 – 2.7.6

- 2.7.1** Use Gauss–Jordan elimination to find the inverse of $\begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$.
- 2.7.2** Verify your answer to Problem 1 by checking $AA^{-1} = I_2$ directly.
- 2.7.3** Is $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ invertible? Justify using any *one* of the equivalent conditions in Theorem 2.7.7.
- 2.7.4** If A, B are both invertible $n \times n$ matrices, simplify $(AB)^{-1}B$ as much as possible using Theorem 2.7.4.
- 2.7.5** Find the inverse of $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ using Gauss–Jordan elimination, showing every step.
- 2.7.6** (*Challenge*) If A is invertible and $A^2 = A$, show that $A = I_n$. (*Hint*: multiply both sides of $A^2 = A$ by A^{-1} .)

4 §2.8 Determinants

Definition 2.8.2 & 2.8.3 – Determinant, Minor, Cofactor

For a square matrix A , the **(i, j) -minor** A_{ij} is the determinant of the submatrix left after deleting row i and column j ; the **(i, j) -cofactor** is $C_{ij} = (-1)^{i+j} A_{ij}$. The **determinant** is defined recursively: $\det[a] = a$ for a 1×1 matrix, and for larger A ,

$$\det(A) = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A(1|j)) = \sum_{j=1}^n a_{1j} C_{1j}$$

(expansion along the first row).

In Plain English

For a 2×2 matrix this collapses to the familiar $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$. For 3×3 , expanding along

row 1 gives $\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$. In general, the determinant is a single number that packages an enormous amount of information about A – most importantly, whether A is invertible at all.

Definition 2.8.6 – Singular / Non-singular

A is **singular** if $\det(A) = 0$, and **non-singular** if $\det(A) \neq 0$.

Theorem 2.8.7 – Effect of Row Operations on the Determinant

- (1) Swapping two rows **negates** $\det(A)$.
- (2) Multiplying a row by c **multiplies** $\det(A)$ by c .
- (3) A row of all zeros forces $\det(A) = 0$.
- (4) Replacing a row by itself plus c times another row **does not change** $\det(A)$.
- (5) Two equal rows force $\det(A) = 0$.

In Plain English

This is where determinants become computationally useful: instead of the recursive cofactor definition (which is $O(n!)$ – catastrophically slow for large n), row-reduce A to upper triangular form using $R_{kj}(c)$ moves (which by (4) don't change the determinant at all), tracking any row swaps (each flips the sign by (2)) or row scalings (each multiplies by that scalar). The determinant of a triangular matrix is just the product of its diagonal entries – so the whole computation becomes as fast as Gaussian elimination itself, $O(n^3)$ instead of $O(n!)$.

Remark 2.8.8 – Expand Along Any Row (or Column)

You don't have to expand along row 1 – expanding along *any* row k , $\det(A) = \sum_{j=1}^n (-1)^{k+j} a_{kj} \det(A(k|j))$, gives the same answer. In practice, always expand along the row or column with the most zeros – each zero entry kills an entire cofactor computation for free.

Definition 2.8.10 – Adjoint

The **adjoint** of A , denoted $\text{Adj}(A)$, is the *transpose* of the cofactor matrix: $\text{Adj}(A) = [b_{ij}]$ with $b_{ij} = C_{ji}$.

Theorem 2.8.12 – The Adjoint Formula for the Inverse

$A \cdot \text{Adj}(A) = \det(A) \cdot I$. Consequently, if $\det(A) \neq 0$,

$$A^{-1} = \frac{1}{\det(A)} \text{Adj}(A).$$

Worked Example – determinant, adjoint, and inverse, all at once

Reusing $A = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$ from §2.7. Expanding along row 1: $\det(A) = 2 \det \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} - 0 + 1 \cdot \det \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = 2(1) - 0 + 1(1) = 3$. Since $\det(A) = 3 \neq 0$, A is non-singular (consistent with it being invertible in §2.7). Computing all nine cofactors and transposing gives

$$\text{Adj}(A) = \begin{pmatrix} 1 & 1 & -1 \\ -1 & 2 & 1 \\ 1 & -2 & 2 \end{pmatrix}, \quad A^{-1} = \frac{1}{3} \text{Adj}(A) = \begin{pmatrix} 1/3 & 1/3 & -1/3 \\ -1/3 & 2/3 & 1/3 \\ 1/3 & -2/3 & 2/3 \end{pmatrix},$$

exactly matching the Gauss–Jordan computation in §2.7 – two completely different methods, same answer.

Theorem 2.8.15, Corollary 2.8.16, Theorem 2.8.17

$\det(AB) = \det(A)\det(B)$ for any square matrices of the same order. A is invertible if and only if $\det(A) \neq 0$. And $\det(A) = \det(A^t)$.

Theorem 2.8.18 – Cramer’s Rule

If $\det(A) \neq 0$, the unique solution of $Ax = b$ (n equations, n unknowns) is

$$x_j = \frac{\det(A_j)}{\det(A)} \quad (j = 1, \dots, n),$$

where A_j is A with its j -th column replaced by b .

Worked Example – Cramer’s rule

Solve $Ax = b$ for $A = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$, $b = \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix}$ (same A as above, $\det(A) = 3$). Replacing each column of A by b in turn:

$$\det(A_1) = \det \begin{pmatrix} 4 & 0 & 1 \\ 2 & 1 & 0 \\ 3 & 1 & 1 \end{pmatrix} = 3, \quad \det(A_2) = \det \begin{pmatrix} 2 & 4 & 1 \\ 1 & 2 & 0 \\ 0 & 3 & 1 \end{pmatrix} = 3,$$

$$\det(A_3) = \det \begin{pmatrix} 2 & 0 & 4 \\ 1 & 1 & 2 \\ 0 & 1 & 3 \end{pmatrix} = 6.$$

So $x_1 = 3/3 = 1$, $x_2 = 3/3 = 1$, $x_3 = 6/3 = 2$: the solution is $(x_1, x_2, x_3) = (1, 1, 2)$. (Quick check: $2(1) + 0(1) + 1(2) = 4$ ✓, $1(1) + 1(1) + 0(2) = 2$ ✓, $0(1) + 1(1) + 1(2) = 3$ ✓.)

Problems 2.8.1 – 2.8.6

2.8.1 Compute $\det \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$.

2.8.2 Compute $\det \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix}$ by cofactor expansion along the first row.

2.8.3 Recompute the determinant in Problem 2 by expanding along a different row or column instead, and check you get the same answer.

2.8.4 Is $\begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$ singular or non-singular?

2.8.5 Find $\text{Adj}(A)$ and A^{-1} for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ using the determinant/adjoint formula, and check against direct 2×2 inverse computation.

2.8.6 (*Challenge*) Use Cramer’s rule to solve $x + y = 5$, $2x - y = 1$, then check your answer using ordinary substitution.

Part 2 Wrap-Up

What You Now Know

- **Rank:** the number of nonzero rows after row reduction, and how Theorem 2.5.7 shows every matrix reduces to an identity block plus zeros.
- The **full** existence/uniqueness theorem (Theorem 2.6.1): comparing $\text{rank}(A)$ against $\text{rank}([A \ b])$ and n tells you, with no exceptions, whether $Ax = b$ has no solution, a unique solution, or infinitely many.
- **Invertible matrices:** uniqueness of the inverse, the “socks and shoes” rule $(AB)^{-1} = B^{-1}A^{-1}$, and the big six-way equivalence theorem (invertible $\Leftrightarrow$ full rank $\Leftrightarrow$ row-equivalent to $I \Leftrightarrow$ product of elementary matrices $\Leftrightarrow Ax = 0$ only trivial $\Leftrightarrow Ax = b$ always solvable), plus the Gauss–Jordan method for actually computing A^{-1} .
- **Determinants:** cofactor expansion, how row operations affect $\det(A)$, the adjoint formula $A^{-1} = \frac{1}{\det(A)} \text{Adj}(A)$, $\det(AB) = \det(A)\det(B)$, and Cramer’s rule for solving $Ax = b$ directly via determinants.

Self-Check Before Chapter 3

You should be able to, without looking anything up:

1. Compute the rank of a 3×3 matrix by row reduction.
2. Given ranks of A and $[A \ b]$, state immediately how many solutions $Ax = b$ has.
3. Compute a 3×3 inverse by *either* Gauss–Jordan elimination *or* the determinant/adjoint formula (and know both methods must agree).
4. State Cramer’s rule and use it on a small system.

If any of these feel shaky, revisit that box above – Chapter 3 (vector spaces) leans on rank and invertibility constantly, right from its first page.

That finishes Chapter 2. Whenever you’re ready, ask for Chapter 3 – Finite Dimensional Vector Spaces – in the same format.